

Manzil JEE 2026

Chemistry (Practice Sheet)

Structure of Atom

Time duration - 120 min.

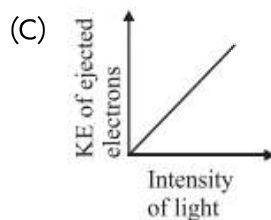
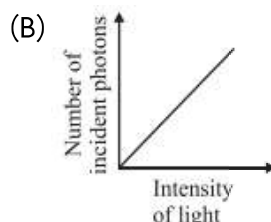
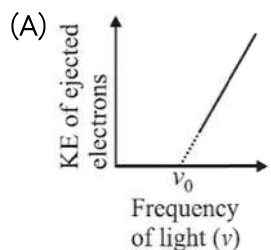
- Q1** Rutherford's experiment which established the nuclear model of the atom, used a beam of
(A) β -particles, which impinged on a metal foil and got absorbed.
(B) γ -rays, which impinged on a metal foil and ejected electrons.
(C) helium atoms, which impinged on a metal foil and got scattered.
(D) helium nuclei, which impinged on a metal foil and got scattered.
- Q2** The speed of electron revolving in the fourth orbit of a hydrogen-like atom or ion is 1094 km/s . The atom or ion is
(A) H
(B) He^+
(C) Li^{2+}
(D) Be^{3+}
- Q3** The number of possible spectral lines in Brackett series in hydrogen spectrum, when the electrons present in the ninth excited state return to the ground state, is
(A) 36 (B) 45
(C) 5 (D) 6
- Q4** For which transition in H-atom, the amount of energy released will be maximum?
(A) $n = 4$ to $n = 2$
(B) $n = 5$ to $n = 2$
(C) $n = 2$ to $n = 1$
(D) $n = 7$ to $n = 2$
- Q5** The magnetic moment order is correctly given in
(A) $\text{Fe}^{2+} < \text{Fe}^{3+} < \text{Mn}^{4+} < \text{Cr}^{3+}$
(B) $\text{Cr}^{3+} = \text{Mn}^{4+} < \text{Fe}^{2+} < \text{Fe}^{3+}$
(C) $\text{Cr}^{3+} = \text{Mn}^{4+} < \text{Fe}^{3+} < \text{Fe}^{2+}$
(D) $\text{Fe}^{2+} < \text{Cr}^{3+} < \text{Fe}^{3+} < \text{Mn}^{4+}$
- Q6** The energy of an excited H-atom is -1.51 eV . Angular momentum of e^- in the given orbit will be
(A) $3 h/\pi$
(B) $3 h/2\pi$
(C) $2 h/\pi$
(D) h/π
- Q7** The distance of closest approach of an α -particle fired towards a nucleus with momentum 'P' is r . What will be the distance of closest approach when the momentum of the α -particle is $2P$?
(A) $2r$
(B) $4r$
(C) $r/2$
(D) $r/4$
- Q8** A bulb of 40 W is producing a light of wavelength 620 nm with 80% of efficiency then the number of photons emitted by the bulb in 20 seconds are
(A) 2×10^{18}
(B) 10^{18}
(C) 10^{21}
(D) 2×10^{21}



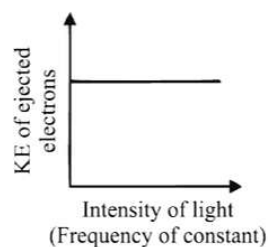
- Q9** Bohr's radius of 2nd orbit of Be^{3+} is equal to that of
 (A) 4th orbit of hydrogen
 (B) 2nd orbit of He^+
 (C) 3rd orbit of Li^{2+}
 (D) First orbit of hydrogen

- Q10** An electron revolves round Li^{2+} nucleus at a distance of $1.587 \text{ \AA}$. The speed of electron should be
 (A) $2.188 \times 10^6 \text{ m/s}$
 (B) $6.564 \times 10^6 \text{ m/s}$
 (C) $7.293 \times 10^5 \text{ m/s}$
 (D) $7.293 \times 10^6 \text{ m/s}$

- Q11** Which of the following is an incorrect graphical representation based on photoelectric effect?



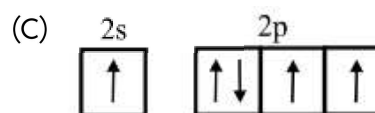
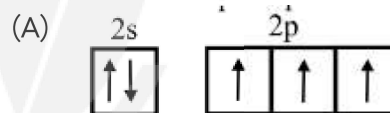
(D)



- Q12** Uncertainty in the position of an electron (mass = $9.1 \times 10^{-31} \text{ kg}$) moving with a velocity 300 m/s accurate up to 0.001% will be
 (A) $5.76 \times 10^{-3} \text{ m}$
 (B) $1.92 \times 10^{-2} \text{ m}$
 (C) $3.84 \times 10^{-3} \text{ m}$
 (D) $19.2 \times 10^{-4} \text{ m}$

- Q13** The radius of an orbit of Be^{+3} ion is $0.529 \text{ \AA}$. The velocity of electron in this orbit will be
 (A) $0.545 \times 10^6 \text{ m/s}$
 (B) $2.18 \times 10^6 \text{ m/s}$
 (C) $4.36 \times 10^6 \text{ m/s}$
 (D) $1.09 \times 10^6 \text{ m/s}$

- Q14** Which of the following configuration is violating Pauli's exclusion principle?



(D) (B) and (C)

- Q15** Which of the following statements is false?
 (A) The energy of red photon is more than the energy of violet photon
 (B) The momentum of photon is inversely proportional to its wavelength
 (C)



The energy of a photon is inversely proportional to its wavelength

- (D) The particle nature of electromagnetic radiations is able to explain the photoelectric effect

Q16 Consider the following statements:

- (a) Electron density in the XY plane in $3d_{x^2-y^2}$ orbital is zero
 (b) Electron density in the XY plane in $3d_{z^2}$ orbital is zero.
 (c) $2s$ orbital has one nodal surface
 (d) for $2p_z$ orbital, XY is the nodal plane.
 Which of these are incorrect statements?

- (A) a & c
 (B) b & c
 (C) Only b
 (D) a, b

Q17 Certain sun glasses having small amount of $AgCl$ incorporated in the lenses, on exposure to light of appropriate wavelength turns to Gray color to reduce the glare following the reactions



If the heat of reaction for the decomposition of $AgCl$ is 248 kJ mol^{-1} , what maximum wavelength is needed to induce the desired process?

- (A) $4.83 \times 10^{-7} \text{ m}$
 (B) $2.45 \times 10^{-7} \text{ m}$
 (C) $4.83 \times 10^{-8} \text{ m}$
 (D) $7.83 \times 10^{-7} \text{ m}$

Q18 The energy of I, II and III energy levels of a certain atom are E , $4E/3$ and $2E$ respectively. A photon of wavelength λ is emitted during a transition from III to I. What will be the wavelength of emission for transition II to I?

(A) $\lambda/2$

- (B) λ
 (C) 2λ
 (D) 3λ

Q19 Among the isotopes of all the elements (radioactive as well as non-radioactive), the packing fraction is maximum for

- (A) ${}_1\text{H}^1$
 (B) ${}_6\text{C}^{12}$
 (C) ${}_1\text{H}^3$
 (D) ${}_{26}\text{Fe}^{56}$

Q20 What is the frequency of the second line of the Paschen series in the spectrum of He^+ ion?

- (A) $\frac{64 R.C}{225}$
 (B) $\frac{64R}{225}$
 (C) $\frac{225}{64R}$
 (D) $\frac{225C}{64R}$

Q21 Which orbital is represented by $\psi_{4,2,0}$?

- (A) $4d$
 (B) $3d$
 (C) $4p$
 (D) $4s$

Q22 The circumference of the third orbit of He^+ ion is $x \text{ m}$. The de-Broglie wavelength of electron revolving in this orbit will be

- (A) $\frac{x}{3} \text{ m}$
 (B) $3x \text{ m}$
 (C) $\frac{x}{9} \text{ m}$
 (D) $9x \text{ m}$

Q23 The speed of electron revolving around H^- nucleus is $0.547 \times 10^6 \text{ m/s}$. The distance of electron from the nucleus is

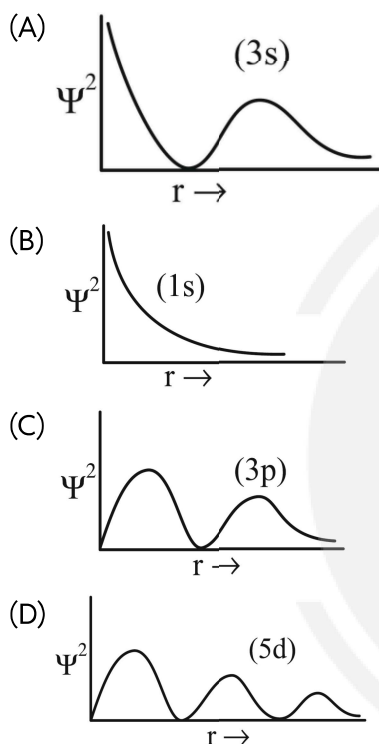
- (A) $2.116 \text{ \AA}$ (B) $4.761 \text{ \AA}$
 (C) $8.464 \text{ \AA}$ (D) $0.529 \text{ \AA}$

Q24 Which of the following statement is not correct?



- (A) $2p$ orbital has no spherical nodes
 (B) The opposite lobes of a p orbital have opposite sign whereas opposite lobes of d orbital have the same sign.
 (C) The number of radial nodes in $3s$ and $4p$ orbital is not equal.
 (D) All p orbitals have only one planar node.

Q25 Which of the given graph below for radial probability density (Ψ^2) v/s distance from nucleus is incorrectly represented?



- Q26** The number of α -particles deflected by $\text{Ca}(20)$ is 40. Find the number of α -particles deflected by $\text{Zn}(30)$?
 (A) 100 (B) 9
 (C) 90 (D) 17

Q27 A single electron species in energy level (orbit number n) with energy X was provided with excess of energy so that it jumps to higher energy level with energy Y . If it can emit radiations of six different wavelengths on de-

excitement between these two energy levels, then the correct relation is

- (A) $\frac{X}{Y} = (n-1)^2$
 (B) $\frac{X}{Y} = 1 + \frac{3}{n}$
 (C) $\sqrt{\frac{X}{Y}} = 1 + \frac{3}{n}$
 (D) $\frac{X}{Y} = 1 + \frac{n}{3}$

Q28 Which of the following statement(s) is incorrect?

- (A) The total spectral lines obtained from a single line during Zeeman effect is $(2l+1)$.
 (B) In the Lyman series as the energy liberated during transition increases then the distance between the spectral lines goes on decreasing.
 (C) The highest probability of finding an electron in $1s$ orbital of hydrogen atom is in the vicinity of the circumference.
 (D) The highest probability of finding an electron in $1s$ orbital of hydrogen atom is exactly at the middle between nucleus and circumference.

Q29 Electronic transition in He^+ ion takes from n_2 to n_1 shell such that:

$$2n_2 + 3n_1 = 18 \quad \dots(i)$$

$$n_2 - n_1 = 4 \quad \dots(ii)$$

Then what will be the total number of photons emitted when electrons transit to n_1 shell?

- (A) 21 (B) 15
 (C) 20 (D) 10

Q30 The value of four quantum number for the last electron of atom of element X are $n = 7, l = 1, m = +1$ and $s = +1/2$ or $-1/2$ and value of spin magnetic momentum for element X is zero. Element X has two isotopes (I) ${}^A_Z X$, (II) ${}^B_Z X$. (Given: $B - A = B - 2Z = 18$, where A and B are atomic masses and Z is atomic number). The value of A and B respectively are



- (A) 118 and 136 (B) 218 and 236
(C) 236 and 254 (D) 226 and 244

- Q31** In two H atoms X and Y the electrons move around the nucleus in circular orbits of radius r and $4r$ respectively. The ratio of the times taken by them to complete one revolution is:
(A) 1 : 4 (B) 1 : 2
(C) 1 : 8 (D) 2 : 1
- Q32** The work function of sodium metal is 4.41×10^{-19} J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electron will be $\text{---} \times 10^{-21}$ J. ($h = 6.63 \times 10^{-34}$ J s; $c = 3 \times 10^8$ m/s)
- Q33** The number of pairs having same electronic configuration is
(Fe^{3+} , Mn^{2+}); (Sc^{2+} , Ti^{3+}); (V^{2+} , Ti^{+}); (Cu^{2+} , Ni^{2+}); (Zn^{2+} , Cu^{+})
- Q34** Element (X) has K-shell configuration $(n - 5)s^2$ and total number of electrons in outermost, penultimate and pre-penultimate shell are 2, 8 and 23 respectively, then the number of unpaired electrons in the ground state of (X) is
- Q35** The bond energy of H-H bond is 79.2×10^{-19} J/molecule. What is the largest wavelength (in nm) of electromagnetic radiation needed to dissociate one H_2 molecule? Assume that one photon may dissociate only one molecule. (Use: $h = 6.6 \times 10^{-34}$ Js; $c = 3 \times 10^8 \times \text{ms}^{-1}$).
- Q36** The number of correct statements from the following ____
(a) For 1s orbital, the probability density is maximum at the nucleus.
(b) For 2s orbital, the probability density first increases to maximum and then decreases

sharply to zero.

- (c) Boundary surface diagrams of the orbitals encloses a region of 100% probability of finding the electron.
(d) p and d-orbitals have 1 and 2 angular nodes respectively.
(e) Probability density of p-orbital is zero at the nucleus.

- Q37** The atomic masses of He and Ne are 4 and 20 a.m.u., respectively. The value of the de Broglie wavelength of He gas at -73°C is "M" times that of the de Broglie wavelength of Ne at 727°C . M is:
- Q38** Which state of the triply ionized Beryllium (Be^{3+}) has the same orbit radius as that of the ground state of hydrogen atom?
- Q39** The Schrodinger wave equation for hydrogen atom is $\Psi_{2s} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r_0}{a_0}\right) e^{-\frac{r_0}{a_0}}$, where a_0 is Bohrs radius. If the radial node is $2s$ be at r_0 , then the value of $\frac{r_0}{a_0}$ is
- Q40** The approximate radius of a H-atom is 0.05 nm, and that of proton is 1.5×10^{-15} m. Assuming both the hydrogen atom and the proton to be spherical, fraction of the space in an atom of hydrogen that is occupied by the nucleus is $p \times 10^{-15}$. Calculate value of p.
- Q41** $\text{N} \xrightarrow{+e^-} \text{N}^-$; the number of INCORRECT set of quantum number for incoming electron in the given process is:
(i) $n \rightarrow 2$, $l \rightarrow 0$, $m_l \rightarrow 0$, $m_s \rightarrow +1/2$
(ii) $n \rightarrow 2$, $l \rightarrow 0$, $m_l \rightarrow 1$, $m_s \rightarrow -1/2$
(iii) $n \rightarrow 1$, $l \rightarrow 1$, $m_l \rightarrow 1$, $m_s \rightarrow +1/2$
(iv) $n \rightarrow 2$, $l \rightarrow 1$, $m_l \rightarrow 0$, $m_s \rightarrow -1/2$
(v) $n \rightarrow 2$, $l \rightarrow 1$, $m_l \rightarrow -1$, $m_s \rightarrow +1/2$
(vi) $n \rightarrow 2$, $l \rightarrow 1$, $m_l \rightarrow 1$, $m_s \rightarrow -1/2$



(vii) $n \rightarrow 2, \quad l \rightarrow 2, \quad m_l \rightarrow 1, \quad m_s \rightarrow -1/2$

(viii)

$n \rightarrow 2, \quad l \rightarrow 1, \quad m_l \rightarrow -2, \quad m_s \rightarrow +1/2$

Q42 A point source of light of power 3.2 milli watts emit energy 5eV . If the source is located at a distance of 0.8 m from the centre of metallic sphere of work function 3eV and radius 8 mm. Number of photoelectrons emitted in 1sec if efficiency is 1% is $x \times 10^y$ in significant number, then find value of 'y' ($1\text{eV} = 1.6 \times 10^{-19} \text{ J}$)

Q43 The wavelength of the spectral line, when the electron in the hydrogen atom undergoes a transition from the energy level 4 to energy level 2 is _____ nm. (Nearest Integer)
($R_H = 109678 \text{ cm}^{-1}$)

Q44 Not considering the electronic spin the degeneracy of the second excited state of H-atom is 'x'. Degeneracy of the second excited state of H^- ion is 'y'. And number of angular node in $d_{x^2-y^2}$ orbital is 'z'. Calculate the value of $\frac{x}{y} + z$?

Q45 Ratio of frequency of revolution of electron in second excited state of He^+ and second stationary state of H is $\frac{2^x}{3^y}$. What is the value of $x + y$?

Q46 An electron in Li^{2+} ions makes a transition from higher state n_2 to lower state $n_1 = 6$. The emitted photons is used to ionize an electron in H-atom from 2nd excited state. The electron on leaving the H-atom has a de-Broglie wavelength $\lambda = 12.016 \text{ \AA}$. Find the value of n_2 .

Note: Use

$$(12.016)^2 = \frac{150 \times 144}{13.6 \times 11}, \quad \lambda_A = \sqrt{\frac{150}{\text{KE}_{\text{eV}}}}$$

Q47 If Z_1 is the number of electrons in g sub shell, Z_2 is the azimuthal quantum number of an atomic

orbital of g sub shell, Z_3 is the minimum principal quantum number of an atomic orbital of g sub shell, Z_4 is the number of angular nodes in an atomic orbital of the g sub shell and Z_5 is the number of elements expected in the g block of the periodic table then find the value of $Z_1 + Z_2 + Z_3 + Z_4 + Z_5$.

Q48 In a sample of hydrogen atoms, all the atoms are in a particular excited state. If the emission spectrum of this sample has only 4 spectral lines in the visible region, then the total number of spectral lines in IR region possible from that state is _____.

Q49 An element undergoes a reaction as shown
 $x + 2e^- \rightarrow x^{-2}$
Energy released = 30.87 eV/atom. If the energy released, is used to dissociate 4g of H_2 molecules equally into H^+ and H^* where H^* is excited state of H atoms where the electron travels in orbit whose circumference equal to four times its de-Broglie's wavelength. Determine the minimum number of moles of x that would be required. Given: IE of H = 13.6 eV/atom, bond energy of H_2 = 4.526 eV/molecules

Q50 In a sample of H-atom electrons make a transition from 5th excited state to the ground state, producing all possible types of photons, then the number of lines in the infrared region is _____



Answer Key

Q1 (D)
Q2 (B)
Q3 (D)
Q4 (C)
Q5 (B)
Q6 (B)
Q7 (D)
Q8 (D)
Q9 (D)
Q10 (A)
Q11 (C)
Q12 (B)
Q13 (C)
Q14 (B)
Q15 (A)
Q16 (D)
Q17 (A)
Q18 (D)
Q19 (A)
Q20 (A)
Q21 (A)
Q22 (A)
Q23 (C)
Q24 (C)
Q25 (A)

Q26 (C)
Q27 (C)
Q28 (D)
Q29 (D)
Q30 (C)
Q31 (C)
Q32 222
Q33 3
Q34 5
Q35 25
Q36 3
Q37 5
Q38 2
Q39 2
Q40 27
Q41 5
Q42 9
Q43 486
Q44 5
Q45 8
Q46 12
Q47 49
Q48 6
Q49 2
Q50 6

